Pre: Think-aloud User Study on GenAI for Software Engineering Tasks

This pre-survey takes approximately 4–6 minutes to complete. Your responses help us understand participants' baseline backgrounds and perspectives, including demographics, prior exposure to Generative AI and attitudes toward GenAI tools, and knowledge assessment of the software engineering (SE) concepts.

Please let the researcher know if you have any questions.

* Indicates required question

1. Email *

Demographics

2. Age *

Mark only one oval.

- 18-24
- 25-34
- 35-44
- 45-54
- 55+
- Prefer not to answer

3. Gender *

Mark only one oval.

- Male
- Female
- Non-binary
- Prefer not to answer
- Other: _____

4. Major (Field of Study) *

_____

5. Highest degree you earned *

Mark only one oval.

- High School or below
- Undergradute
- Master
- PhD

6. Current role(s) *

Check all that apply.

- [ ] Undergraduate Student
- [ ] Master's Student
- [ ] PhD Student
- [ ] Software Developer
- [ ] Project Manager
- [ ] Other: _____

7. Years of software engineering industrial experience *

Mark only one oval.

- ◯ None
- ◯ 3-6 months
- ◯ 7-12 months
- ◯ 1-2 years
- ◯ 3-5 years
- ◯ 5+ years

8. Which development tasks have you experienced in software engineering projects *

Check all that apply.

- [ ] Requirement Analysis
- [ ] User Interface Design
- [ ] Architecture Design
- [ ] Implementation (e.g., add a feature, code review, debug)
- [ ] Testing
- [ ] Deployment
- [ ] Maintenance
- [ ] N/A
- [ ] Other: _____

Prior Experience and General Opinions on GenAI

9. Which GenAI tools have you used for **non-SE tasks**? (Select all that apply) *

Check all that apply.

- [ ] ChatGPT
- [ ] GitHub Copilot
- [ ] Google Gemini
- [ ] Claude
- [ ] Google AI Studio
- [ ] Cursor
- [ ] Perplexity AI
- [ ] Haven't used any
- [ ] Other: _____

10. What do you primarily use GenAI for **non-SE tasks**? *

_____

_____

_____

_____

_____

11. Which GenAI tools have you used for **SE tasks** (Select all that apply) *

Check all that apply.

- [ ] ChatGPT
- [ ] GitHub Copilot
- [ ] Google Gemini
- [ ] Claude
- [ ] Google AI Studio
- [ ] Cursor
- [ ] Perplexity AI
- [ ] Haven't used any
- [ ] Other: _____

12. Which of the following **SE tasks** have you used GenAI tools for? (Select all that apply) *

Check all that apply.

- [ ] Concepts, Frameworks, and Languages Learning
- [ ] Requirement Analysis
- [ ] User Interface Design
- [ ] Architecture Design
- [ ] Implementation (e.g., Code Generation, Code Review, Code Refactoring)
- [ ] Testing
- [ ] Deployment
- [ ] Maintenance
- [ ] N/A
- [ ] Other: _____

13. How often do you use GenAI overall? *

Mark only one oval per row.

	Multiple times per day	Daily	Several times per week	Weekly	Monthly	Rarely	Never
SE tasks	◯	◯	◯	◯	◯	◯	◯
Non-SE tasks	◯	◯	◯	◯	◯	◯	◯

14. Overall, how effective do you find GenAI tools for SE tasks?

Mark only one oval.

	1	2	3	4	5	
Extr	◯	◯	◯	◯	◯	Extremely Effective

15. Which of the following input modes do you **commonly use** when interacting with GenAI tools? (Select all that apply) *

Check all that apply.

- [ ] Text (typing prompts)
- [ ] Audio/Voice
- [ ] Screen sharing
- [ ] File upload (e.g., PDF, Word, Excel)
- [ ] Image upload
- [ ] Video upload
- [ ] I have not used GenAI tools before
- [ ] Other: _____

16. Which of the following multimodal features have you used in GenAI tools? *

Check all that apply.

- [ ] Text-based chat
- [ ] Interactive voice assistant
- [ ] Live screen sharing
- [ ] Multimodal search (AI ingests diverse inputs, e.g., text, image, audio, video)
- [ ] Cross-model understanding (e.g., image and video understanding, audio transcription)
- [ ] Cross-model generation (e.g., text-to-image, text-to-audio, text-to-video)
- [ ] I have not used GenAI tools before
- [ ] Other: _____

17. How often do you verify the accuracy of content generated by GenAI tools? *

Mark only one oval.

- ◯ Always
- ◯ Often
- ◯ Sometimes
- ◯ Seldom
- ◯ Never
- ◯ Other: _____

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Knowledge Assessment of the Software Engineering Concepts

18. How familiar are you with [concept] in software engineering? *

Mark only one oval per row.

	Not at all familiar	Slightly familiar	Moderately familiar	Very familiar	Extremely familiar
Nielsen's Usability Heuristics	◯	◯	◯	◯	◯
Software Requirements Specification	◯	◯	◯	◯	◯
Functional Requirements	◯	◯	◯	◯	◯
Use Case	◯	◯	◯	◯	◯
Use Story	◯	◯	◯	◯	◯

19. What is the primary purpose of a Software Requirements Specification (SRS) document? *

Mark only one oval.

- ◯ To document the internal code architecture and algorithms
- ◯ To outline the project's marketing and sales plan
- ◯ To serve as the agreed baseline before development begins
- ◯ To log bugs discovered during testing

20. Per IEEE guidance, which of these is considered a quality attribute of a good SRS? *

Mark only one oval.

- Unverifiable, to avoid limiting the design
- Written only in code, not natural language
- Ambiguous, so it can be interpreted flexibly
- Complete and consistent, with no conflicts

21. What is the primary focus of a use case in software engineering? *

Mark only one oval.

- The programming language used to build the system
- What the system does from the user's perspective
- The specific hardware the system runs on
- How the internal code is structured

22. In a use case, what does an 'actor' represent? *

Mark only one oval.

- A specific function in the source code
- A step in the testing checklist
- A table in the system's database
- A person or system interacting externally

23. What is the main purpose of a user story in software development? *

Mark only one oval.

- To capture a short description of desired functionality
- To serve as a complete, final technical specification
- To describe the CI/CD deployment pipeline
- To formally document the system's non-functional requirements

24. Which of these is written in a typical user story format? *

Mark only one oval.

- "Section 4.2: Password Reset Functionality Specification Document"
- "Use Case: Reset Password. Actor: Customer. Preconditions: account exists..."
- "The system shall process 1,000 transactions per second, minimum"
- "As a customer, I want to reset my password, so that I can regain access to my account."

25. Which of the following is an example of a functional requirement? *

Mark only one oval.

- The system must be available 99.9% of the time
- The system shall allow login with username and password
- The system must respond to any request within 2 seconds
- The system must comply with GDPR data-retention rules

26. What do functional requirements primarily describe? *

Mark only one oval.

- The specific services and behaviors the system provides
- The budget and timeline for development
- The hardware needed to run the system
- Quality attributes like speed, security, or reliability

27. What is the purpose of usability heuristics (e.g., Nielsen's 10)? *

Mark only one oval.

- To specify exact pixel dimensions for every UI element
- To provide broad rules for evaluating usability
- To calculate the total cost of development
- To define the database schema for an application

28. Which of the following is one of Nielsen's 10 usability heuristics? *

Mark only one oval.

- Continuous integration
- Object-oriented encapsulation
- Visibility of system status
- Database normalization

Knowledge Assessment of the Software Engineering Concepts

29. How familiar are you with [concept] in software engineering? *

Mark only one oval per row.

	Not at all familiar	Slightly familiar	Moderately familiar	Very familiar	Extremely familiar
Nielsen's Usability Heuristics	◯	◯	◯	◯	◯
Software Requirements Specification	◯	◯	◯	◯	◯
Functional Requirements	◯	◯	◯	◯	◯
Use Case	◯	◯	◯	◯	◯
Use Story	◯	◯	◯	◯	◯

30. What is the primary purpose of a Software Requirements Specification (SRS) document? *

Mark only one oval.

- ◯ To define the visual branding and marketing materials for launch
- ◯ To define what the system must do and guide its design
- ◯ To act as a permanent contract that can never be revised after signing
- ◯ To document defects found once the product is live

31. Which of the following would typically belong inside an SRS? *

Mark only one oval.

- A list of all Git commit messages from the sprint
- The team's daily stand-up notes from every week's sprint
- A functional requirement describing what the system must do
- The full source code written for the unit tests

32. What is the primary focus of a use case in software engineering? *

Mark only one oval.

- The exact syntax of the programming language used to build the system
- The estimated cost and timeline of the project
- A goal-oriented interaction between a user and the system
- The physical servers and network topology the system runs on

33. For an ATM's 'Withdraw Cash' use case, which of the following would be listed as a precondition? *

Mark only one oval.

- The customer's account balance is updated after withdrawal
- The customer receives their cash from the machine
- The bank designates the customer as the primary actor
- The customer's card has been inserted and PIN verified

34. What is the main purpose of a user story in software development? *

Mark only one oval.

- To specify the exact user interface layout and pixel measurements
- To provide a short, user-focused description of a feature
- To serve as a binding legal agreement with the client
- To document every edge case and exception before development starts

35. A user story reads: "As a user, I want to reset my password, so that I can regain access to my account." Which of the following is an example of acceptance criteria for this story? *

Mark only one oval.

- "As a user, I want to reset my password, so that I can regain access to my account."
- "Given a valid email, when the user requests a reset link, then an email is sent within 1 minute.
- "The database schema shall include a table called password_reset_tokens for storing reset requests."
- "The team has estimated that this story is worth approximately 3 story points."

36. Which of the following is an example of a functional requirement? *

Mark only one oval.

- The system must encrypt all sensitive data while at rest
- The system should load all pages within 500 milliseconds
- The system shall allow users to reset their password via email
- The system must maintain 99.99% uptime during business hours

37. Which pairing correctly matches each requirement to its type? *

Mark only one oval.

- "Checkout page must load in under 1 second" is functional; "Users can apply a discount code" is non-functional
- "Users can apply a discount code" is functional; "Checkout page must load in under 1 second" is non-functional
- Both statements are functional requirements
- Both statements are non-functional requirements

38. What is the purpose of usability heuristics (e.g., Nielsen's 10)? *

Mark only one oval.

- To calculate the return on investment for a redesign
- To provide general principles that help evaluate interface usability
- To define the exact color palette a product must use
- To specify the programming framework a team should adopt

39. A website shows a plain spinning loader with no message or progress indicator during a 30-second file upload. Which usability heuristic is being violated? *

Mark only one oval.

- Aesthetic and minimalist design
- Consistency and standards
- Visibility of system status
- Recognition rather than recall

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